

Activity 13

Marching Orders—*Programming Languages*

Summary

Computers are usually programmed using a “language,” which is a limited vocabulary of instructions that can be obeyed. One of the most frustrating things about programming is that computers always obey the instructions to the letter, even if they produce a crazy result. This activity gives students some experience with this aspect of programming.

Curriculum Links

- ✓ English: Interpersonal Listening

Skills

- ✓ Giving and following instructions.

Ages

- ✓ 7 years and up

Materials

You will need:

- ✓ Cards with pictures such as the ones shown on the next page.

Each student will need:

- ✓ Pencil, paper and ruler

Marching Orders

Introduction

Discuss whether it would be good if people followed instructions exactly. For example, what would happen if you pointed to a closed door and said, “Go through that door”?

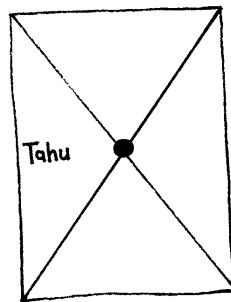
Computers work by following lists of instructions, and they do exactly what the instructions say—even if they don’t make sense!

Demonstration Example

See if the students can draw the picture from these instructions.

1. Draw a dot in the centre of your page.
2. Starting at the top left-hand corner of the page rule a straight line through the dot finishing at the bottom right hand corner.
3. Starting at the bottom left-hand corner of the page rule a line through the dot, finishing at the top right hand corner.
4. Write your name in the triangle in the centre of the left-hand side of the page.

The result should look something like this:



Activities

Choose a student and give them an image (such as the examples on page 124). The student describes the picture for the class to reproduce. The students can ask questions to clarify the instructions. The object is to see how quickly and accurately the exercise can be completed.

Repeat the exercise, but this time the students are not allowed to ask questions. It is best to use a simpler image for this exercise, as the students can get lost very quickly.

Now try the exercise with the instructing student hidden behind a screen, without allowing any questions, so that the only communication is in the form of instructions.

Point out that this form of communication is most like the one that computer programmers experience when writing programs. They give a set of instructions to the computer, and don't find out the effect of the instructions until afterwards.

Have the students draw a picture and write down their own instructions. Try them out in pairs or as a whole class.

Variations

1. Write instructions to construct a paper dart.
2. Write instructions on how to get to a mystery location around the school using such instructions as "Go forward x metres", "turn left" (90 degrees), and "turn right" (90 degrees).

Students should test and refine their instructions until they have the desired effect.

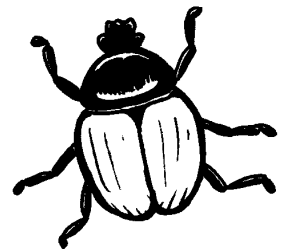
3. Blind Game. Blindfold a student and have the other students direct them around the room.

What's it all about?

Computers operate by following a list of instructions, called a program, that has been written to carry out a particular task. Programs are written in languages that have been specially designed, with a limited set of instructions, to tell computers what to do. Some languages are more suitable for some purposes than others.

Regardless of what language they use, programmers must become adept at specifying *exactly* what they want the computer to do. Unlike human beings, a computer will carry out instructions to the letter even if they are patently ridiculous.

It is important that programs are well written. A small error can cause a lot of problems. Imagine the consequences of an error in the program of a computer in a space shuttle launch, a nuclear power plant, or the signals on a train track! Errors are commonly called “bugs” in honour (so it is said) of a moth that was once removed (“debugged”) from an electrical relay in an early 1940s electronic calculating machine.



The more complex the program, the more errors there are likely to be. This became a major issue when the USA was working on the Strategic Defence Initiative (“Star Wars”) program, a computer controlled system that was intended to form an impenetrable defence against nuclear attack. Some computer scientists claimed that it could never work because of the complexity and inherent unreliability of the software required. Software needs to be tested carefully to find as many bugs as possible, and it wouldn’t be feasible to test this system since one would have to fire missiles at the United States to be sure that it worked!

